

SS-RESfM vs. RESfM: Why the Paper Reads as a Critique — and Why It Isn't One

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1. Intent vs. reading

The paper’s *intent* is constructive extension: it uses RESfM’s architecture, code lineage, and benchmark family throughout, and its best arm is literally RESfM’s network with a different (self-supervised) training signal. But it contains four empirical findings that, read cold — e.g. by an anonymous reviewer, or by the RESfM authors meeting it as a submission rather than hearing it first from us — recalibrate the RESfM paper’s claims downward. This is the source of the political sensitivity in the venue decision; it is about the findings’ valence, not the paper’s tone.

2. The four findings that carry critical valence

1. **The reproduction gap.** RESfM’s published MegaDepth number (0.169) is not reachable from the released recipe: from-scratch faithful training lands at 0.34–0.39. We decomposed the gap into three layers — eval-environment ($0.169 \rightarrow 0.209/0.217$ under a pinned stack; solver nondeterminism), checkpoint provenance (the released checkpoint is not recipe-reproducible), and the training-seed lottery (0.19–0.5 across seeds). Politely stated: *the headline number reflects a checkpoint that cannot be regenerated from the paper’s recipe.*
2. **The seed lottery.** Five-seed bands show supervised RESfM is bimodal: on BlendedMVS, and on Olsson *even in-domain* (2.03 ± 1.86 ; good on 3/5 seeds). The cross-generalization grid (58 cells) shows the supervised Olsson clean-benchmark collapse is training-domain-invariant — a property of the recipe, not of any training pool. A single-seed paper (theirs reports seed 20, one run) sitting on a bimodal distribution invites the reading that the published results are the lucky mode.
3. **The label ceiling.** The COLMAP-derived supervision labels degrade sharply above $\sim 15\%$ contamination (label recall collapses to ~ 0.3 , coverage to $\sim 55\%$). At the 60% end of the axis, the supervised model does no better than no outlier mechanism at all (23.4 vs. 23.1). The supervision that gives RESfM its name earns its keep only in the ~ 25 – 43% contamination band.
4. **The chicken-and-egg dependency.** Generating RESfM’s training labels requires a successful COLMAP reconstruction (4px reprojection cutoff against COLMAP-triangulated 3D) of the very scenes the method is meant to make tractable — a circular dependency our unsupervised mechanisms exist to remove.

3. The other side: why “critique” overstates it

- The paper **corroborates RESfM’s own Table 7** — our findings extend a limitation the original authors themselves reported.

- It **adopts their architecture, benchmark family, and evaluation protocol** wholesale; every improvement is built on their design decisions being right.
- The strongest arm is **their network with a different training signal** — the message is “the architecture is right; the supervision is the fragile part; here is how to fix it.” That is a compliment to the design and a continuation of the lab’s research line.
- Fadi’s own follow-up work already gestures at some of these limits; the findings are a natural next chapter, not an ambush.
- The reliability findings cut *both* ways: our self-supervised arms are shown with the same 5-seed honesty, including where they lose (e.g. supervised wins the 25–43% band; Strecha exception).

4. Why this matters for the venue decision

Findings 1 and 2 are about **reproducibility of a specific published result**, and no framing fully softens that for the authors of that result. Whether Ronen and Fadi read the paper as “our student strengthened our method and found real limits” (likely, if they hear it from us first, with the 3-page summary) or as something sharper (possible, if they first meet it as an anonymous CVPR submission reviewed by their community) is decided by *sequencing*, not by wording. This is the entire reason the advisor email is the critical path of the submission plan (see `VENUES_acceptance_chances.pdf`, Section 4).

5. Framing rules already adopted in the drafts

- Findings-first doctrine: the findings *are* the paper, not the delta over RESfM; the 5-seed benchmark capability is presented as the contribution.
- Table-7 corroboration is cited explicitly, positioning the work as confirming and extending the authors’ own observation.
- Reproduction-gap language is decomposition-based (environment / provenance / seed), never insinuating; the seed-lottery discussion includes the “couldn’t one just pick a good seed?” rebuttal to preempt the adversarial reading.
- Both regimes of ESfM* and the oracle-cleaning ceiling are reported, which *strengthens* the parent line’s framing (outlier handling matters more than clean tracks).